

Query Dual Dynamics Delta (QD³)

WRITE

$(\mathbf{k}_t, \mathbf{v}_t)$

Query importance

$$h_j(\mathbf{k}_t) = \frac{\sum_i a_{j,i} (\mathbf{u}_{j,i}^T \mathbf{k}_t)^2 / s_j}{\sum_i a_{j,i} (\mathbf{u}_{j,i}^T \mathbf{k}_t)^2 / s_j}$$

Clipped rate

$$\tilde{\eta}_t = \min \{ \eta(1 + \beta h_j), c \}$$

Delta value memory

$$\begin{aligned} \hat{\mathbf{v}}_t &= \mathbf{M}_{t-1} \mathbf{k}_t \\ \mathbf{e}_t &= \mathbf{v}_t - \hat{\mathbf{v}}_t \\ \mathbf{M}_t &\leftarrow \mathbf{M}_{t-1} + \tilde{\eta}_t \mathbf{e}_t \mathbf{k}_t^T \end{aligned}$$

read history

Low-rank query trace

$$\begin{aligned} \mathbf{z}_j &= \mathbf{U}_{j-1}^T \mathbf{q}_j \\ \mathbf{r}_j &= \mathbf{q}_j - \mathbf{U}_{j-1} \mathbf{z}_j \\ \mathbf{U}_j &\leftarrow \mathbf{U}_{j-1} + \eta_u \mathbf{r}_j \mathbf{z}_j^T \\ \mathbf{a}_j &\leftarrow \lambda \mathbf{a}_{j-1} + \mathbf{z}_j^{\odot 2} \end{aligned}$$

QUERY

$\mathbf{q}_j$

$\mathbf{M}_t$

$\beta=0 \Rightarrow$ Plain Delta

Recall

$$\begin{aligned} \hat{\mathbf{v}}_j &= \mathbf{M}_t \mathbf{q}_j \\ \text{value state unchanged} \end{aligned}$$

OUTPUT

$\hat{\mathbf{v}}_j$

same $\mathbf{q}_j$